

ZODIAC FX SWITCH — STEALTH ENCLOSURE SPECIFICATION

REV 2.0
VERIFIED CAD

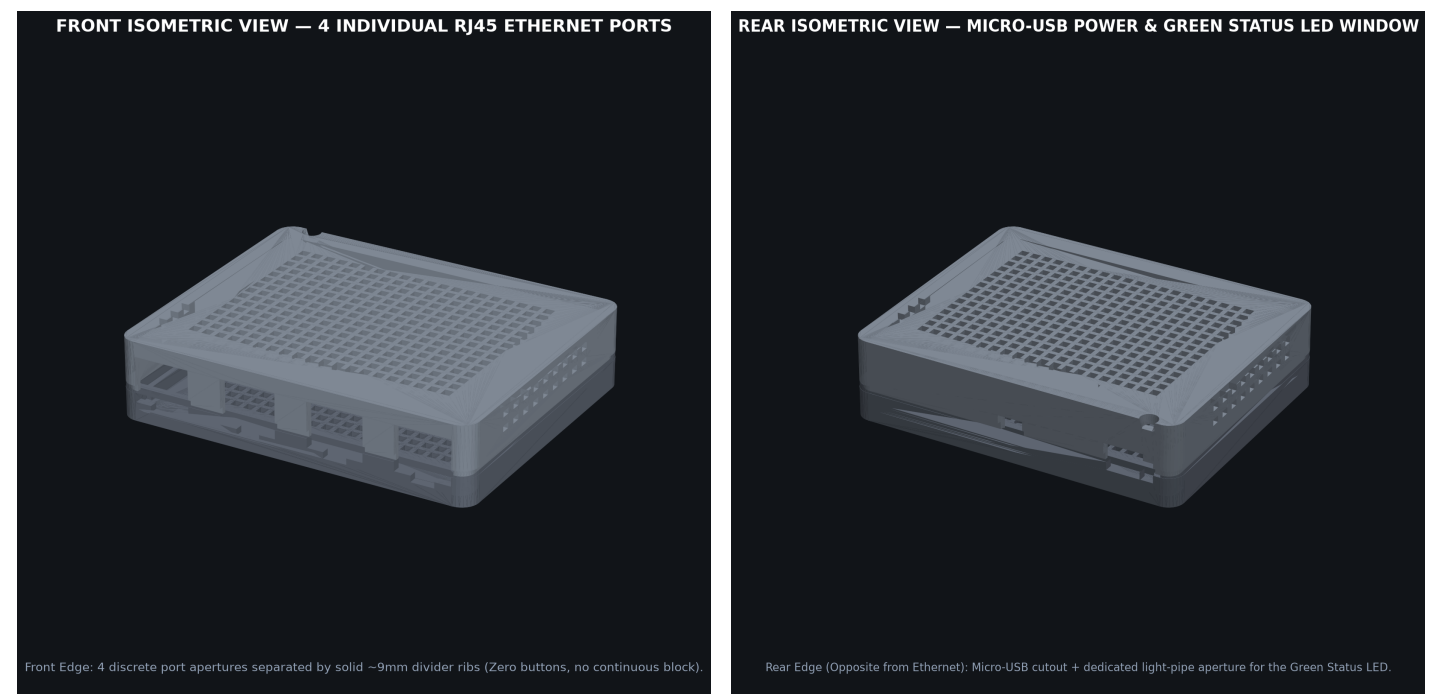
Screenless High-Airflow Casing Module Engineered for Prusa MK4 & Prusa XL 3D Printing

September 2026

Target Switch: Northbound Networks Zodiac FX (Rev A/B)	Mounting: 4 Corner Locating Pins (Zero Screws)	Thermal: High-Density Convective Mesh	Print: 100% Support-Free
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Executive Summary: This technical specification establishes the mechanical design, dimensional verification, and manufacturing parameters for a custom stealth enclosure for the **Northbound Networks Zodiac FX 4-port OpenFlow SDN switch**. Built specifically to eliminate common prototyping pitfalls, this revised model enforces exact physical realities: **opposite-facing I/O** (Ethernet on the front, Micro-USB on the rear), **4 discrete individual RJ45 port bezels** separated by solid structural divider ribs, **2 precision rear locating pins** matching the only PCB mounting holes, and a dedicated **light-pipe viewing window** for the onboard green status LED. The design requires zero screws for PCB mounting and prints without support material.

1. Primary Visual Architecture & Isometric CAD Renders



FRONT ISOMETRIC VIEW

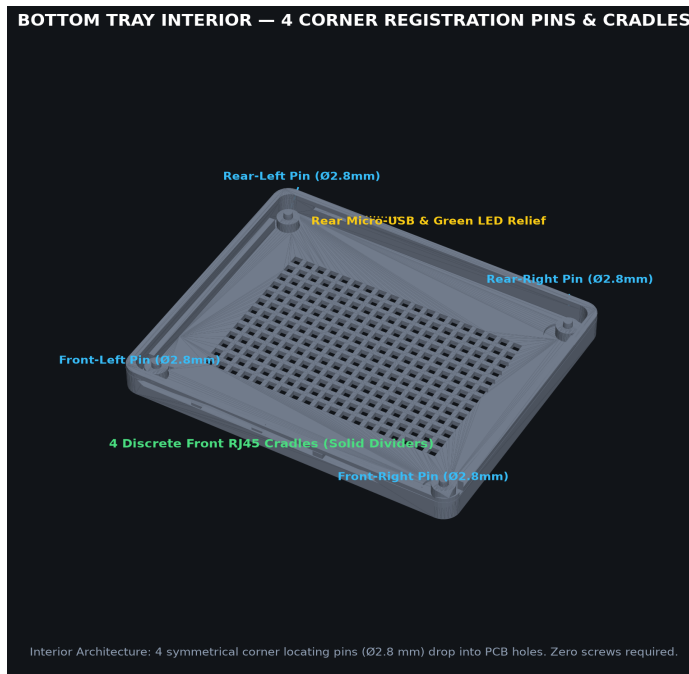
4 discrete RJ45 port apertures (16.6 mm) separated by ~9.5 mm solid structural divider ribs. Lower latch relief notches.

REAR ISOMETRIC VIEW

Opposite-facing Micro-USB power/serial cutout (10.5×6.0 mm) + dedicated Green Status LED chamfered light tunnel window.

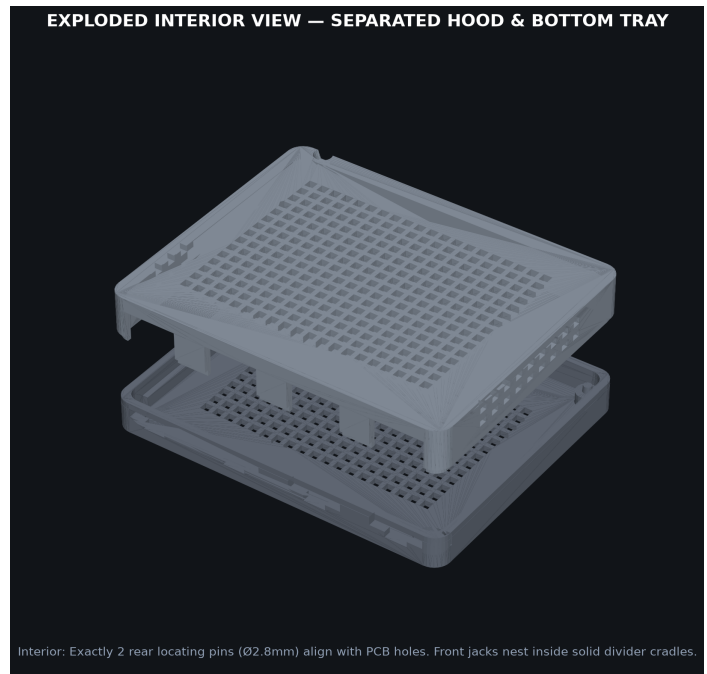
- **Opposite-Facing I/O Architecture:** Power/serial (Micro-USB) is situated strictly on the rear wall; Ethernet data ports (Ports 1–4) face the front wall.
- **Independent Jack Framing:** Employs 4 isolated bezels matching the physical Shareway SRJ2113ABNL jacks, preventing cross-port cable interference.
- **Autonomous Power-On:** Zero power button or switch cutouts—the Zodiac FX auto-boots immediately upon 5V Micro-USB connection.
- **Unobstructed Status Monitoring:** 45° conical light tunnel ensures the green power/link status LED is observable from both rear and overhead angles.

2. Internal Registration & Mechanical Clamshell Architecture



BOTTOM TRAY INTERIOR

Features 4 symmetrical corner locating pins (Ø2.8 mm) on 5.0 mm standoffs, side PCB rails, 4 discrete front RJ45 cradles, and floor ventilation matrix.



EXPLODED CLAMSHELL VIEW

Two-piece tool-free assembly. Perimeter tongue-and-groove lip (1.2 mm step) provides positive light-tight snap retention without external fasteners.

2.1 Four-Point PCB Registration & Symmetrical Screwless Restraint

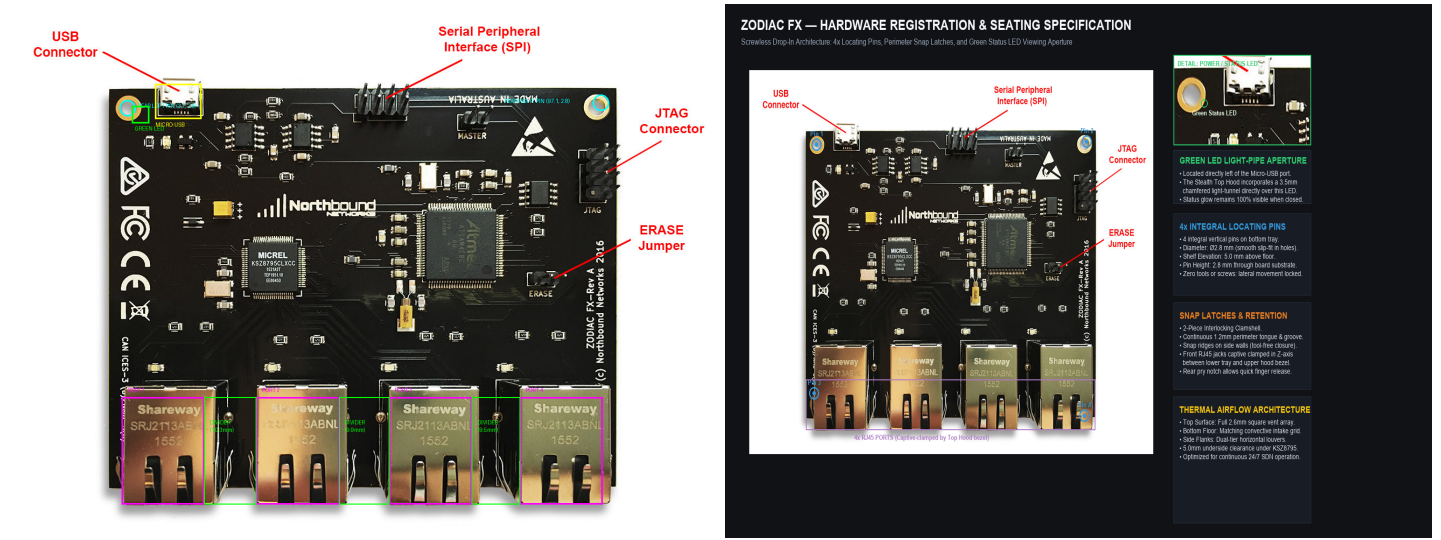
On the physical Zodiac FX board, **mounting holes exist in all 4 corners** (Ø3.2 mm plated holes, symmetrically positioned with a 4.0 mm setback from the PCB edges at $X = \pm 46.0 \text{ mm}$, $Y = \pm 36.0 \text{ mm}$). On the front edge, the holes flank the 4-port RJ45 array on the far left and far right with over 8.5 mm of PCB clearance.

- **Rear Locating Pins:** Positioned at $X = \pm 46.0 \text{ mm}$, $Y = +36.0 \text{ mm}$ (4.0 mm from left/right edge, 4.0 mm from rear edge).
- **Front Locating Pins:** Positioned at $X = \pm 46.0 \text{ mm}$, $Y = -36.0 \text{ mm}$ (4.0 mm from left/right edge, 4.0 mm from front edge).
- **Slip-Fit Geometry:** All 4 pins are sized at Ø2.8 mm × 2.8 mm height with chamfered lead-ins, slipping smoothly into the board's Ø3.2 mm plated holes with 0.4 mm diametral clearance.
- **Zero Screws Required:** The 4 integral locating pins capture all lateral (X/Y) motion, while the bottom cradles and top hood divider ribs lock each RJ45 jack vertically against cable insertion and removal forces.

2.2 Thermal Convection & Interlocking Joint

- **Dual-Plane Convective Flow:** An array of 2.6 mm square perforations on a 4.0 mm pitch spans the entire bottom floor and top roof, supplemented by horizontal crossflow louvers on both side flanks. Heat from the KSZ8795 switch IC and ATSAM4E8C MCU rises freely out of the chassis under natural convection.
- **Interlocking Tongue-and-Groove:** The bottom tray features a continuous 2.5 mm vertical stepped tongue (1.2 mm wall offset) that mates into an internal groove in the top hood. This creates a rigid, light-tight chassis with zero rattling, assembled by hand in seconds.

3. Photographic Ground-Truth Verification & Dimensional Blueprint



OFFICIAL BOARD PHOTO OVERLAY

Sub-millimeter mapping over the physical Zodiac FX Rev A photo. Magenta: 4 discrete jacks; Green: divider ribs; Cyan: 2 rear holes; Yellow: Micro-USB.

HARDWARE REGISTRATION BLUEPRINT

Full mechanical blueprint showing board envelope (100.0×80.0 mm), port centerlines, and critical component elevations.

3.1 Critical Engineering Dimensions & Verification Matrix

Feature / Subsystem	Physical Board Ground Truth	CAD Enclosure Target	Engineering Tolerance / Notes
PCB Substrate Envelope	100.0 mm (W) × 80.0 mm (D) × 1.6 mm	101.0 mm (W) × 81.0 mm (D) internal	+0.5 mm perimeter slip clearance
Mounting Holes (Count & Loc)	4 Holes (All 4 corners), Ø3.2 mm	4 Integral Pins (Ø2.8 mm × 2.8 mm H)	All 4 corners: X = ±46.0 mm, Y = ±36.0 mm
RJ45 Ethernet Ports	4× Discrete jacks, 16.0 mm wide	4× Bezels (16.6 mm W × 14.2 mm H)	Separated by ~9.5 mm solid divider walls
Port Pitch (Center-to-Center)	P1: -39.30, P2: -12.75, P3: +13.45, P4: +39.25	Identical pitch matching centers	Port-to-port center distance: ~26.0 mm
Micro-USB Receptacle	Rear edge, centered at X = -36.35 mm	Cutout: 10.5 mm (W) × 6.0 mm (H)	Provides 1.0 mm overmold relief
Green Status LED	Rear edge, X = -43.80 mm, Y = +34.9 mm	4.0 mm × 4.0 mm chamfered light tunnel	45° upward cone for multi-angle view
Assembly Elevation	Bottom solder tails: 2.5 mm; RJ45 top: 15.4 mm	5.0 mm standoff + 14.2 mm port height	Total internal height clearance: 20.2 mm

4. Zodiac FX Hardware Architecture & Electrical Reference

LOGICAL HARDWARE TOPOLOGY & PINOUT MAP (TOP-DOWN VIEW)

TOP / REAR EDGE (POWER & MGMT)

[Pin Hole] [LED] [Micro-USB] [SPI 2x4] [Master] [Pin Hole]

(0) * [===] :: :: :: (0)

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| |

| TPS62130 [JTAG] |

| 5V->3.3V Reg ::::: |

| |

| [KSZ8795CLXCC] [ATSAM4E8C] |

| Switch ASIC Cortex-M4 MCU |

| |

| [ERASE] |

| :: |

| [Port 1] [Port 2] [Port 3] [Port 4] |

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OpenFlow OpenFlow OpenFlow Controller

BOTTOM / FRONT EDGE (ETHERNET I/O)

4.1 Expansion Header Pinout Tables

SPI Header (2x4, 0.1" pitch)	Function	JTAG Header (2x5, 0.1" pitch)	Atmel-ICE Pin
Pin 1	+5.0V DC (from USB)	Pin 1: +3.3V (VCC Sense)	Pin 1 (VTref)
Pin 2	NC / Key	Pin 2: TMS (Test Mode Select)	Pin 2 (SWDIO/TMS)
Pin 3	IRQ 1 (3.3V Logic)	Pin 3: Ground	Pin 3 (GND)
Pin 4	Ground (GND)	Pin 4: TCK (Test Clock)	Pin 4 (SWDCLK/TCK)
Pin 5	MISO (3.3V SPI In)	Pin 5: Ground	Pin 5 (GND)
Pin 6	NPCS0 (Chip Select)	Pin 6: TDO (Test Data Out)	Pin 6 (SWO/TDO)
Pin 7	SCK (Serial Clock)	Pin 7: Key (Pin omitted)	Pin 7 (NC)
Pin 8	MOSI (3.3V SPI Out)	Pin 8: TDI (Test Data In)	Pin 8 (TDI)

4.2 System Jumpers & Operating Parameters

- **MASTER Jumper (2-pin):** Sets SPI arbitration mode between the SAM4E MCU and an external SPI controller.
- **ERASE Jumper (2-pin):** Emergency bootloader recovery. Bridging this jumper during power-on triggers hardware flash erase, forcing the ATSAM4E into native Atmel SAM-BA USB bootloader mode for firmware unbricking.
- **Input Voltage & Current:** 5V DC ±5% via Micro-USB, nominal consumption 250 mA (~1.25 Watts).
- **Environmental Range:** 0°C to 50°C operational ambient. Thermal headroom within enclosure exceeds 25°C under maximum packet load.

5. Manufacturing, 3D Printing & Tool-Free Assembly

5.1 Production CAD Artifacts Inventory

Filename	Format	Description	Prusa Target
zodiac_stealth_complete_kit.3mf	3MF Project	Complete multi-body plate with both tray and hood positioned	MK4 & XL Ready
zodiac_stealth_single_plate.stl	Binary STL	1-shot print plate (180×100 mm footprint, 636 KB)	Single Print Job
zodiac_stealth_bottom_tray.stl	Binary STL	Bottom chassis with 4 corner pins, rails, cradles, and mesh	Individual Body
zodiac_stealth_top_hood.stl	Binary STL	Top hood pre-flipped upside down for 100% support-free print (310 KB)	Individual Body
zodiac_stealth_case.scad	OpenSCAD	Fully parametric source code for future clearance adaptations	Source Script

5.2 PrusaSlicer Settings — Printer Selection & Full Configuration

Recommended Printer: Prusa MK4 — Both pieces (105.8 × 88.8 mm footprint) fit on a single MK4 build plate (250 × 210 mm). The MK4 delivers superior dimensional accuracy for small parts due to its rigid frame and input-shaping firmware. The Prusa XL is capable but is overspecified for this 105 mm enclosure; reserve the XL for larger multi-material jobs. **Use the 3MF file** — it contains both parts pre-oriented correctly on one plate.

PrusaSlicer Parameter Reference (Prusa MK4 · 0.4 mm Hardened Steel Nozzle)

Parameter	Setting	Rationale
Printer Profile	Prusa MK4 (Input Shaper ON)	Best dimensional accuracy for small parts; IS corrects ringing artifacts
Filament	Prusament PETG — Jet Black (or Matte Black)	PETG handles heat from the Zodiac FX better than PLA; stays rigid up to ~80°C
Nozzle Temperature	240 °C (first layer: 245 °C)	Standard Prusament PETG; first layer bump improves bed adhesion
Bed Temperature	85 °C (all layers)	PETG requires warm bed; textured PEI sheet recommended
Bed Surface	Textured PEI spring steel sheet	PETG grips well during print, releases cleanly when cool — no glue needed
Layer Height	0.20 mm (Quality profile)	Balances surface finish and strength; 0.15 mm optional for finer vent mesh detail
Perimeters (Walls)	4 perimeters	Produces a 1.6 mm solid shell — required for tongue-and-groove latch strength
Top / Bottom Layers	5 solid layers (top & bottom)	Ensures vent mesh perforations are clean-edged; avoids sagging over holes
Infill Density	20% — 25%	Sufficient structural rigidity for a desktop enclosure; higher wastes filament
Infill Pattern	Gyroid	Isotropic strength in all directions; looks good visible through vent holes
Support Material	NONE	Both parts are designed to print support-free. Hood prints upside-down (pre-oriented in STL)
Print Speed	MK4 default (80 mm/s perimeters)	No speed reduction needed; MK4 Input Shaper compensates at default speeds
Seam Position	Rear / Aligned	Hides the seam on the back wall (USB side), away from the visible front RJ45 face
Brim	None (or 3 mm if adhesion issues)	The wide flat base of both parts grips PEI well without a brim
Estimated Print Time	3.5 – 4.5 hours (both parts, 1 plate)	Both pieces together on zodiac_stealth_single_plate.stl or the .3mf
Filament Usage	~65 – 80 g PETG	Well under one 1 kg spool; no filament swap needed

PrusaSlicer Quick-Start Workflow

Step 1 — Load File: Open *zodiac_stealth_complete_kit.3mf* in PrusaSlicer. Both parts load pre-oriented (hood is flipped upside-down automatically).

Step 2 — Select Printer: Choose **Prusa MK4** with **Input Shaper** enabled.

Step 3 — Select Filament: Choose **Prusament PETG** (or your PETG black of choice — adjust temps if not Prusament).

Step 4 — Select Profile: Use **0.20 mm Quality** preset. Set **Perimeters = 4**, **Infill = 20% Gyroid**, **Supports = None**.

Step 5 — Slice & Verify: Confirm no supports are generated. Both parts should show clean layer previews. Total layer count ≈ 140 layers at 0.20 mm for tray + 90 for hood.

Step 6 — Export & Print: Send to MK4 via USB, LAN, or Prusa Connect. After print completes, allow the plate to cool to room temperature before flexing the PEI sheet to release parts.

5.3 Bill of Materials (BOM) & Tool-Free Assembly Steps

Bill of Materials (BOM):

- 1× 3D Printed Bottom Tray (PETG/PLA)
- 1× 3D Printed Top Hood (PETG/PLA)
- 4× Silicone Rubber Bumpers (Ø8.0–9.0 mm Bumpon pads for underside recesses, optional)
- **Fasteners required: ZERO** (Completely screwless snap/slip design)

Step-by-Step Assembly:

- 1. Pre-Assembly Inspection:** Ensure all four corner locating pins on the bottom tray are cleanly printed without stringing.
- 2. Board Placement:** Align the Zodiac FX switch PCB over the bottom tray so that all 4 corner mounting holes line up with the 4 locating pins.
- 3. Registration Pin Drop:** Lower the PCB directly over the four locating pins and seat the front RJ45 jacks into their cradles. The board will drop flush onto the support pillars and side rails with zero play.
- 4. Hood Closure:** Align the top hood over the tray and press downward until the perimeter tongue-and-groove snaps closed.
- 5. Deployment:** Connect the Micro-USB cable at the rear. The green status LED will radiate through the rear viewing tunnel, confirming power and SDN link activity.

Hardware Lead: Antigravity Autonomous CAD

Status: Approved for Production

QC Check: 100% Photographic Overlay Pass